RAJALAKSHMI ENGINEERING COLLEGE

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K2

Started on	Friday, 11 September 2026, 9:22 AM
State	Finished
Completed on	Friday, 11 September 2026, 9:33 AM
Time taken	10 mins 50 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1		a	g	**g**	**t**	**a**	**b**
s2	**g**	x	**t**	**x**	**a**	**y**	**b**

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab azb	2

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <string.h>
#include <stdlib.h>
int max(int a, int b) {
    return (a > b) ? a : b;
}
int longestCommonSubsequence(char *s1, char *s2) {
    int m = strlen(s1);
    int n = strlen(s2);
    int **dp = (int **)malloc((m + 1) * sizeof(int *));
    for (int i = 0; i <= m; i++) {
        dp[i] = (int *)malloc((n + 1) * sizeof(int));
    }
    for (int i = 0; i <= m; i++) {
        for (int j = 0; j <= n; j++) {
            if (i == 0 || j == 0) {
                dp[i][j] = 0;
            } else if (s1[i - 1] == s2[j - 1]) {
                dp[i][j] = dp[i - 1][j - 1] + 1;
            } else {
                dp[i][j] = max(dp[i - 1][j], dp[i][j - 1]);
            }
        }
    }
   int result = dp[m][n];
    for (int i = 0; i <= m; i++) {
        free(dp[i]);
    }
    free(dp);
   return result;
}
int main() {
    char s1[1005], s2[1005];
    if (scanf("%s %s", s1, s2) == 2) {
        printf("%d\n", longestCommonSubsequence(s1, s2));
```

```
36      }
37      return 0;
38  }
```

	Input	Expected	Got	
✔	aab azb	2	2	✔
✔	ABCD ABCD	4	4	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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